Moving surgery centers and uniform first-order stability

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The previous I52 proof established first-order stability of selected shrinking-surgery metrics inside explicit TT or Bianchi slices. The missing passage from arbitrary nearby metrics to a fixed slice is not merely an unproved estimate. The construction below shows that such a passage, with uniformly bounded metric distortion, can fail on the actual recovery family.

The obstruction is a marked geometric variable. A harmonic map on the smooth negatively curved limit is fixed by its homotopy class, whereas an isotopy on the resolved manifold must preserve the exceptional class and hence its concentration point. Moving that point relative to the limiting harmonic map makes the two requirements incompatible.

The positive continuation keeps the centers as parameters. I52's first-order inequality then holds uniformly on compact configuration families, including their conformal and diffeomorphism saturations. It does not assert that every nearby metric admits this parametrization. Continuing past this obstruction, the companion matched-cap proof I53 now removes the gauge and scalar assumptions from the lower first-order bound near selected recovery bases. It does not solve a fixed exact slice.

1. Notation and actual recovery geometry

Let X be a closed complex-hyperbolic surface with $\mathrm{Ric}(g_X) = -g_X$, and let $A = c_1^2(X)$. On

$$M = X\#k\overline{\mathbb{CP}}^2\#\ell(S^1\times S^3), \qquad k \geqslant 1,$$

put $Q = k + 4\ell$, $h = Q/A$, and

$$P(g) = \int |\mathrm{Ric}_g^\circ|^2, \quad S(g) = \int s_g^2, \quad R(g) = \int |\mathrm{Ric}_g|^2.$$

The relative Einstein defect is the same functional as in I40–I52. For a scalar-minus-four metric it is

$$\mathscr{E}(g) = \frac{4P(g)}{R(g)} = \frac{4P(g)}{P(g) + 4V(g)}. \tag{1.1}$$

All classes and actual monopole suppliers are inherited from the same smooth manifolds. No new local solution is substituted for a closed conformal supplier.

We use the actual Burns blowup and two-ended scalar-flat Schwarzschild handle constructions of LeBrun, Theorem 4.3, with negative Yamabe normalization as justified in I52, Section 3. For a fixed X, k, ℓ, their normalized shrinking limits satisfy

$$P(g_t) \longrightarrow 8\pi^2 Q, \qquad V(g_t) \longrightarrow 2\pi^2 A, \qquad \mathscr{E}(g_t) \longrightarrow \frac{4h}{1+h}. \tag{1.2}$$

The last limit is a recovery value, not an assertion about the unrestricted infimum.

For the marked argument it is important to specify the handles. We put the two mouths of each handle in one shrinking background ball. If R_t is its outer radius, d_t the separation of the mouths, r_t their cutoff radius and a_t the donor scale, choose

$$a_t \ll r_t^2, \qquad r_t \ll d_t \ll R_t \longrightarrow 0. \tag{1.3}$$

All clusters are disjoint, and the limiting cluster points are distinct. First select the cluster sizes; then choose the inner scales sufficiently small for the finitely many required energy and normalization errors. This is a fixed-topology diagonal, before taking any large-cover limit.

The whole handle cluster has diameter tending to zero. The Schwarzschild neck has diameter $O(r_t)$, and the remaining cluster has diameter $O(R_t)$. The normalized Yamabe factors have a uniform upper bound from the uniform negative lower bound for the raw scalar curvature. These statements also hold for the Burns regions. The metrics converge smoothly to the background off the finite set of limiting points. Consequently the marked metric spaces converge to the smooth X.

This choice is essential: handles joining two fixed distinct points would instead identify those points in the metric limit. We do not use that different limit here. Clustering does not change the smooth topology, the actual supplier, or the costs in (1.2).

The cutoffs and the exterior estimates of I52 remain available. One uses an enclosing cutoff for each whole cluster and much smaller cutoffs at the two mouths for construction. There is bounded curvature energy per cluster, and the transition errors are chosen arbitrarily small. Thus these clustered bases can satisfy all the selections in I52, Theorem 9, simultaneously with (1.3).

2. Comparing normalization without resolving the cores again

The following elementary observation is useful when the raw metrics agree near the entire singular region.

Lemma 1 (relative Yamabe normalization). Suppose $b_{0,t}$ and $b_{1,t}$ are smooth raw metrics on the same closed manifold, equal outside a fixed regular region K. Let $g_{0,t} = u_t^2 b_{0,t}$ have scalar curvature -4. Assume on a neighborhood of K that

$$u_t \longrightarrow 1 \text{ in } C^2, \qquad b_{i,t} \longrightarrow b_i \text{ smoothly}, \qquad s_{b_i} = -4. \tag{2.1}$$

If $\|b_{1,t} - b_{0,t}\|_{L^\infty(b_{0,t})} \leqslant a$, then their scalar-minus-four Yamabe representatives satisfy

$$\|g_{1,t} - g_{0,t}\|_{L^\infty(g_{0,t})} \leqslant a + o_t(1)(1 + a). \tag{2.2}$$

No lower bound for the size of the cores is needed.

Proof. Put $\tilde{g}_t = u_t^2 b_{1,t}$. It equals $g_{0,t}$ off K. The conformal scalar formula and (2.1) show

$$\delta_t := \|s_{\tilde{g}_t} + 4\|_\infty \longrightarrow 0.$$

For sufficiently small t, this metric has negative scalar curvature. Write its normalized representative as $g_{1,t} = v_t^2 \tilde{g}_t$. At the maximum and minimum of v_t, the scalar equation gives

$$1 - \delta_t/4 \leqslant v_t^2 \leqslant 1 + \delta_t/4. \tag{2.3}$$

The common u_t^2 cancels in the relative pointwise tensor norm, which proves (2.2). QED.

This proof requires only the already established exterior convergence. It does not infer uniform convergence of u_t on every rescaled donor from exterior convergence.

3. A fixed-base Bianchi slice can have unbounded distortion

Our divergence convention is

$$\beta_g q = \delta_g q + \tfrac{1}{2} d\ \mathrm{tr}_g\, q, \qquad (\delta_g q)_j = -\nabla^i q_{ij}.$$

For a diffeomorphism φ, set $q = \varphi^* g_1$. Direct contraction of the connection difference gives

$$\tau(\mathrm{id} : (M, g_0) \to (M, q))^k = -q^{k\ell}(\beta_{g_0} q)_\ell. \tag{3.1}$$

Thus $\beta_{g_0}(\varphi^* g_1 - g_0) = 0$ is exactly the harmonic-map equation for $\varphi : (M, g_0) \to (M, g_1)$, not only its linearization.

Theorem 2 (marked obstruction to a uniform fixed slice). For the balanced actual family used in I40–I52, with $h_j \to 0$ and at least one blowup, there exist scalar-minus-four recovery metrics g_j^0, g_j^1 on the same M_j such that

$$\|g_j^1 - g_j^0\|_{L^\infty(g_j^0)} \longrightarrow 0, \qquad \frac{\mathscr{E}(g_j^0)}{h_j} \longrightarrow 4, \qquad \frac{\mathscr{E}(g_j^1)}{h_j} \longrightarrow 4, \tag{3.2}$$

and every $\varphi \in \mathrm{Diff}_0(M_j)$ satisfying

$$\beta_{g_j^0}(\varphi^* g_j^1 - g_j^0) = 0 \tag{3.3}$$

has bilipschitz constant greater than j, measured from g_j^0 to g_j^1. If no such diffeomorphism exists, the statement has the usual empty-set meaning. The same examples can also be chosen with

$$\|\delta^*_{g_j^0} \beta_{g_j^0}(g_j^1 - g_j^0)\|_2 = o(\sqrt{Q_j}). \tag{3.3a}$$

Thus even a small actual gauge residual does not imply a uniformly controlled exact gauge at the fixed marked base.

The bases g_j^0 may in addition satisfy the selections of I52. In particular there is no automatic uniformly bounded, let alone vanishing-amplitude, exact Bianchi representative relative to those fixed bases under just constant scalar curvature, competitive energy, and C^0 closeness.

Proof: move the smooth background while retaining the marked core. First keep X, k, ℓ fixed. Choose one exceptional point p. In a CH chart about p, choose a local Killing field moving p, multiply it by a cutoff, and let ψ_ϵ be its small-time flow. It is isotopic to the identity, is a genuine local CH isometry on a smaller neighborhood of p, and is the identity near all other surgery clusters and outside a larger chart. Set

$$b_\epsilon = \psi_\epsilon^* g_X.$$

Then $b_\epsilon = g_X$ near every donor and outside a fixed regular annulus K, while

$$\|b_\epsilon - g_X\|_{C^m} \leqslant C_m |\epsilon|, \qquad \psi_\epsilon^{-1}(p) \neq p \quad (\epsilon \neq 0). \tag{3.4}$$

Perform exactly the same marked surgeries at exactly the same points in g_X and b_ϵ. The two raw metrics are identical near all donors, and differ only on K. Normalize them to obtain $g_{0,t}$ and $g_{\epsilon,t}$. Lemma 1 and the recovery limits give, for fixed ϵ,

$$\|g_{\epsilon,t} - g_{0,t}\|_{L^\infty(g_{0,t})} \leqslant C|\epsilon| + o_t(1), \tag{3.5}$$

and (1.2) for both families.

Proof: compactness of a hypothetical bounded-distortion gauge. Suppose for this fixed nonzero ϵ there are $t_n \to 0$ and L-bilipschitz maps $\varphi_n \in \mathrm{Diff}_0(M)$ satisfying (3.3). The collapsing regions described in Section 1 have diameter tending to zero. Uniform convergence on the smooth exterior, or equivalently the elementary compactness of bilipschitz maps between the resulting compact metric spaces, produces a bilipschitz homeomorphism

$$F : (X, g_X) \longrightarrow (X, b_\epsilon). \tag{3.6}$$

The inverse maps converge as well, so this is not merely a continuous map onto a quotient.

The exceptional class at p forces

$$F(p) = p. \tag{3.7}$$

Indeed the exceptional sphere E_p in the source has diameter tending to zero. Its image also does, and $[\varphi_n(E_p)] = [E_p] \neq 0$ in $H_2(M; \mathbb{Z})$. If its limit were a regular point of the target X, it would eventually lie in a contractible coordinate ball and have zero class. If its limit were another exceptional point q, it would lie in one fixed small enclosing topological neighborhood of that cluster, whose image in $H_2(M)$ is $\mathbb{Z}[E_q]$, not $\mathbb{Z}[E_p]$. A whole handle cluster has $H_2 = 0$. These exhaust the points other than p, proving (3.7). We use the entire neighborhoods and their marked homology, not just terminal bubbles.

Proof: the limit is in the identity homotopy class. Collapse maps $\pi_n : M \to X$ induce the usual quotient

$$\pi_1(M) = \pi_1(X) * F_\ell \longrightarrow \pi_1(X).$$

Isotopy of φ_n gives $\pi_n \circ \varphi_n \simeq \pi_n$. Choose generators on loops avoiding the finite source cluster set and F^{-1} of the target cluster set. On these loops the convergence is in the ordinary smooth regions, so the induced outer homomorphism of F on $\pi_1(X)$ is the identity. The CH manifold X is aspherical. Hence F is homotopic to the identity. This argument explicitly kills the free handle generators; it does not identify $\pi_1(M)$ with $\pi_1(X)$.

Proof: the harmonic limit contradicts the mark. By (3.1), each φ_n is harmonic. Away from the finite set consisting of source cluster points and inverse images of target cluster points, both metrics converge smoothly and the differential of φ_n is uniformly bounded. In coordinates the harmonic equation has bounded right-hand side; interior elliptic estimates and bootstrapping give smooth convergence there.

The limit (3.6) is Lipschitz on the smooth compact X. In the weak harmonic equation a cutoff at a missing point has an error bounded by Cr^3: the derivative of the map is bounded and the cutoff has derivative $O(r^{-1})$ on a four-dimensional annulus of volume $O(r^4)$. The equation therefore extends weakly across every missing point. Coordinate elliptic regularity makes F a smooth harmonic map on all of X.

The map $\psi_\epsilon^{-1} : (X, g_X) \to (X, b_\epsilon)$ is an isometry, hence harmonic, and is homotopic to the identity. Strictly negative sectional curvature of b_ϵ makes this harmonic map unique in its homotopy class. One may apply Hartman, Section 1(H), rank-greater-than-one corollary, p.675. Equivalently, geodesic-homotopy energy convexity is strict against the rank-four isometry unless the homotopy is stationary. Thus $F = \psi_\epsilon^{-1}$, contradicting (3.7) and (3.4).

We have proved that for each fixed nonzero ϵ and each finite L, sufficiently small donor scales exclude every such L-bilipschitz gauge.

Proof: the balanced diagonal. Return to X_j, k_j, ℓ_j. Choose $\epsilon_j \to 0$, using uniformly controlled lifted CH charts; if needed choose it even smaller to absorb any chart constant. For each fixed j, ϵ_j, choose the clustered donor scales so small that:

- the just-proved exclusion holds with $L = j$;
- the relative error in (3.5) tends to zero;
- both errors in (1.2) are $o(Q_j)$ in P and small enough in V to give (3.2);
- all the core, exterior, and near-stationarity errors required in I52 are at most a prescribed number tending to zero.

Each list is finite at fixed j, so the same diagonal satisfies all requirements. Since $4/(1 + h_j) \to 4$, (3.2) follows. No assertion about the rate at which the necessary bilipschitz constant diverges is required. The remaining strengthening (3.3a) is proved next and is included in the same diagonal. QED.

3.1 The counterexamples still satisfy the approximate-gauge criterion

This is a positive coverage statement as well as a distinction between an approximate and an exact gauge. Write

$$r_t = u_t^2(b_{\epsilon,t} - b_{0,t}), \qquad \tilde{g}_t = g_{0,t} + r_t, \qquad g_{\epsilon,t} = e^{2f_t}\tilde{g}_t, \qquad w_t = e^{2f_t} - 1.$$

The tensor r_t is supported in the regular annulus K, with $\|r_t\|_{C^2(g_{0,t},K)} \leqslant C|\epsilon|$. Lemma 1 gives $\|f_t\|_\infty \leqslant C\delta_t$, where $s_{\tilde{g}_t} + 4$ is supported in K, has supremum $\delta_t \to 0$, and has L^2 norm $O(\delta_t)$. Also $P(\tilde{g}_t) \leqslant CQ$, since outside K it is the same metric as $g_{0,t}$, while its curvature on K converges smoothly to that of the CH background b_ϵ.

The derivative bounds proved in I52, Lemma 5, now apply on $\tilde{g}_t$:

$$\|\nabla^2 f_t\|_2 \leqslant C\delta_t(1 + \sqrt{Q}), \qquad \|df_t\|_2 \leqslant C\delta_t, \qquad \|df_t\|_4^2 \leqslant C\delta_t^2(1 + \sqrt{Q}). \tag{3.8}$$

Changing the connection in these estimates back to $g_{0,t}$ only costs $C|\epsilon|\|df_t\|_{2,K}$; the two metrics agree off K. Consequently the Hessian of w_t has the same first bound in (3.8).

Put $k_t = g_{\epsilon,t} - g_{0,t} = w_t g_{0,t} + (1 + w_t)r_t$. The exact product rules are

$$\beta_0(wg_0) = dw, \qquad \beta_0(fr) = f\beta_0 r + \Big(\tfrac{1}{2}\operatorname{tr}_0 r\, I - r^\sharp\Big)df.$$

One further derivative therefore gives

$$\|\delta_0^*\beta_0 k_t\|_2 \leqslant C\{|\epsilon| + \delta_t(1 + \sqrt{Q})\}. \tag{3.9}$$

Choose the donor scales also to ensure $\delta_t(1 + \sqrt{Q}) \leqslant |\epsilon|$. With $\epsilon_j \to 0$ and $Q_j \geqslant 1$, equations (3.5) and (3.9) prove (3.3a) and

$$\|k_j\|_\infty \, \|\delta_0^*\beta_0 k_j\|_2 = o(\sqrt{Q_j}). \tag{3.10}$$

Thus these actual examples are already covered by the approximate criterion of I52, Corollary 11. They do not refute that criterion or show that moving centers is the only way to retain coefficient four. What fails is the attempted replacement of a small residual by a uniformly controlled exact fixed-base harmonic slice. If a chosen regular chart or cutoff has constants depending on j, choose the displacement first so that the resulting $C_j|\epsilon_j| \to 0$, and then the donor scales so that $C_j\delta_j(1 + \sqrt{Q_j}) \leqslant$

$C_j|\epsilon_j|$. This is permitted by the fixed-j, fixed-displacement argument and does not change the exact-gauge exclusion.

What the theorem does not say. These are not sub-four metrics, not failures of actual monopole supply, and not counterexamples to I52 or to Yau26. It also does not exclude other useful gauges, changing the background centers, or approximate gauge equations. It rules out a precise proposed reduction to a fixed exact Bianchi slice, even on competitive actual recovery metrics.

4. Retaining the centers gives a positive uniform extension

The obstruction above disappears for its explicit family if the centers are moved with the background. Let $\mathbf{p}$ be the marked centers and local frames in the first construction and $\mathbf{q} = \psi_\epsilon(\mathbf{p})$. The local isometry at each affected center extends over the corresponding Burns donor using the transported frame; the other donors are unchanged. Thus there is a marked diffeomorphism between the two surgery manifolds with the exact raw identity

$$b_{\epsilon,t} = \Psi_\epsilon^* b_t(g_X, \mathbf{q}). \tag{4.1}$$

Negative Yamabe normalization commutes with pullback, by uniqueness, so

$$g_{\epsilon,t} = \Psi_\epsilon^* g_t(g_X, \mathbf{q}). \tag{4.2}$$

This is an exact equivalence, not an approximate gauge assertion. The family defeated a fixed center, not the full family of centers.

We next record the uniform statement which follows from the actual estimates of I52; keeping it uniform is important when the number and positions of the cores vary.

Theorem 3 (uniform moving-center stability). For each j, let K_j be a compact set of marked, pairwise distinct cluster-center configurations and frames in the CH cover X_j, with the fixed counts k_j, ℓ_j. The enclosing balls are disjoint and lie in uniformly controlled CH charts. There are recovery families $g_j^0(\mathbf{p})$, $\mathbf{p} \in K_j$, whose diagonal errors are uniform on K_j, such that the conclusion of I52, Theorem 9 holds for every choice $\mathbf{p}_j \in K_j$, with constants independent of these choices.

In particular, if $a_j = \|k_j^{\mathrm{met}}\|_\infty \to 0$ and k_j^{met} is TT relative to $g_j^0(\mathbf{p}_j)$, then

$$\liminf_j \frac{\mathscr{E}\left(\widehat{g_j^0(\mathbf{p}_j) + k_j^{\mathrm{met}}}\right)}{h_j} \geqslant 4. \tag{4.3}$$

The same holds in the constant-scalar Bianchi version of I52. The exact-gauge quadratic lower bound and equality criterion are uniform as well. The symbol k_j^{met} is a tensor and is distinct from the blowup count k_j.

Proof. For each fixed j, compactness of K_j supplies a positive minimum separation of its distinct cluster centers, not a separation independent of j. Choose enclosing radii below that separation and so small that their fourth powers sum to at most $\varepsilon_j Q_j$, with $\varepsilon_j \to 0$. Within the handle clusters use the further hierarchy (1.3). All local metric and cutoff estimates take place in CH charts. The number of cutoffs is $O(Q_j)$; each has uniformly bounded scale-invariant second-derivative cost.

The donor costs, scalar lower bound, and exterior convergence are uniform over a compact configuration set at fixed j. For completeness, nonuniform exterior convergence would give a sequence of

configurations converging in K_j. After moving their finitely many centers by smooth chart identifications, the capacity and negative-Yamabe uniqueness proof of I52, Lemma 3, applies to the limiting configuration, contradicting that failure. Local elliptic estimates give the required exterior C^2 convergence. The raw and normalized Bach and Ricci-gradient errors are treated in the same finite compact family. A common sufficiently small scale therefore makes their errors at most $\varepsilon_j Q_j$. In particular impose the actual stronger exterior selection $P_{\mathrm{ext}} \leqslant \varepsilon_j^2 Q_j^2/A_j$, as well as $\|\operatorname{grad} P(g_j^0)\|_1 \leqslant \varepsilon_j Q_j$. An exterior bound of only $o(Q_j)$ would not justify that near-stationarity statement.

The constants in the remaining I52 argument use only the universal CH tensor gap, the bounded energy per donor, the disjoint cutoff estimates, and these selected errors. They do not use the minimum separation itself. Hence its graph estimate and square completion give, uniformly in $\mathbf{p}$,

$$P_{\mathrm{final}} \geqslant 8\pi^2 Q_j + c x_j^2 - C a_j^2 Q_j - \varepsilon_j Q_j, \tag{4.4}$$

where x_j is the appropriate norm-preserving identified linearization from I52. The same opposite-chirality conversion proves (4.3). The matching expansion in the competitive case gives the inherited equality criterion $x_j^2 = o(Q_j)$. QED.

There need not be a global trivialization of the entire configuration bundle of smooth manifolds. The assertions are fiberwise and invariant under the marked local identifications. For the one-parameter family in Theorem 2 a contractible chart of configurations suffices.

Corollary 4 (conformal and diffeomorphism saturation). The lower bound (4.3) is unchanged if one takes any orientation-preserving pullback and any positive smooth conformal multiple of its output. Neither the conformal factor nor the diffeomorphism needs a uniform coordinate bound.

Proof. Diffeomorphism invariance is exact. If $\widehat{g}$ is the negative constant-scalar representative of a conformal class and g is any other representative, put

$$\delta = S(g) - S(\widehat{g}) \geqslant 0, \qquad L = S(g) - \frac{(\int s_g)^2}{V(g)} \geqslant 0.$$

The four-dimensional conformal curvature identities give

$$P(g) = P(\widehat{g}) + \delta/12, \qquad R(g) = R(\widehat{g}) + \delta/3.$$

Consequently

$$\mathscr{E}(g) = \frac{\mathscr{E}(\widehat{g})R(\widehat{g}) + \delta/3 + L}{R(\widehat{g}) + \delta/3} \geqslant \min\{\mathscr{E}(\widehat{g}), 1\}. \tag{4.5}$$

Since $h_j \to 0$, this inequality transfers (4.3), including cases of noncompetitive conformal representatives. QED.

Theorems 2–3 belong together: a configuration family is genuinely needed for the proposed uniformly controlled exact-slice description of these explicit C^0-nearby recovery metrics, and the first-order estimate is uniform when that variable is kept. The approximate criterion supplies a separate fixed-base route for this example. We have not proved that this modulated family covers all nearby metrics, nor that every unrestricted near-minimizer lies near it.

5. The intrinsic continuation: matched collars without a global slice

A second route avoids a global harmonic slice. The complete companion proof constructs a full-curvature cap of cost $C(\int |\mathrm{Rm}|^2 + \|g - e\|_\infty^2)$, matching the actual outer collar exactly. Localized DeTurck smoothing supplies an interior smooth band; a fixed-input frozen-collar estimate and critical energy control complete the two matches.

No scalar normalization is needed. Directly, for $\mathscr{E} < 1/2$, Chern–Gauss–Bonnet gives total curvature $O(\chi) = O(A_j)$. In L_j disjoint fixed-ratio bands around each marked blowup, minimum-energy selection gives

$$\sum_i \int_{\mathscr{A}_{i,j}} |\mathrm{Rm}|^2 \leqslant CA_j/L_j = o(Q_j) \quad \text{if } L_j h_j \to \infty. \tag{5.1}$$

The annuli enclose designated blowups, not handles. Leave any energy-bad cores unremoved; their count is $o(Q_j)$ when $L_j h_j \to \infty$. The partial signature identity charges these retained blowups explicitly. It gives

$$\mathscr{E}(g_j) \geqslant \frac{4h_j}{1+h_j} - C/L_j - C(k_j/A_j)a_j^2. \tag{5.1a}$$

This proves the coefficient-four lower bound for arbitrary C^0-nearby smooth competitors on selected long-annulus bases, and computes the infimum in every shrinking C^0 neighborhood. It also allows $o(Q_j)$ wholly unstructured blowups and only mean-square small annular amplitudes.

The remaining all-metric question is whether unrestricted near-minimizers possess enough such marked annuli. The cap itself is no longer the missing step. No global conformal normalization is used to infer zeroth-order closeness.

There is no static curvature-supremum estimate for arbitrary metrics from these inputs. For example, on a fixed local annulus consider

$$g = dx_1^2 + e^{2u(x_1)}dx_2^2 + e^{-2u(x_1)}dx_3^2 + v(x_1)^2 dx_4^2, \qquad v'' = (2r^2 - (u')^2)v, \quad v(0) = 1,\ v'(0) = 0.$$

Its scalar curvature is exactly $-4r^2$. With a compactly supported slab $u = a\psi((x_1 - c)/\ell)$, on a bounded interval,

$$\|g - g_{\mathrm{Eucl}}\|_\infty = O(a + r^2 + a^2/\ell), \qquad \int |\mathrm{Rm}|^2 = O(a^2/\ell^3 + r^4), \qquad \|\mathrm{Rm}\|_\infty \gtrsim a/\ell^2. \tag{5.2}$$

Choose $a = \ell^p$, $3/2 < p < 2$, and $r \to 0$. Then the first two quantities vanish while the last diverges. The estimates follow from the multiply warped curvature formulas and the displayed ODE; v stays positive because $r^2 + a^2/\ell \to 0$. This local test does not disprove a low-energy cap.

Positive-time Ricci smoothing is a possible construction tool, but matching the original collar remains an extra step. Even using an actual Ricci flow rather than DeTurck coordinates, a variable-time graph $g_\theta(x) = g(x, \theta(x))$ creates connection terms

$$B^k{}_{ij} = -g^{k\ell}\{\mathrm{Ric}_{j\ell}\,\theta_i + \mathrm{Ric}_{i\ell}\,\theta_j - \mathrm{Ric}_{ij}\,\theta_\ell\}. \tag{5.3}$$

Its curvature includes $d\theta \nabla \mathrm{Ric}$, $\nabla^2\theta\ \mathrm{Ric}$, and $|d\theta|^2(\partial_t \mathrm{Ric} + \mathrm{Ric} * \mathrm{Ric})$. Pointwise smoothing bounds of size a/t alone give a nonintegrable majorant near the hypersurface where $\theta = 0$. These terms explain why positive-time smoothing alone was insufficient. I53 instead uses a spatially localized flow with the full $\mathscr{L}_{q\mathrm{W}}$ term, controls its integrated energy, and then restores the original collar. It does not use a variable-time graph or assert that the original metric has bounded curvature.

6. Status and classical inputs

The new actual counterexample is registered as X09. The uniform moving-center and conformal extensions are recorded with I52, not counted as several separate breakthroughs.

O11, the all-metric coefficient on the balanced family, remains open. O10, the unrestricted actual weak zero-Weyl endpoint, also remains open. The present theorem neither supplies a closed endpoint escape nor rules one out. The continued endpoint equations are maintained separately in the research record.

Classical inputs used here are the Burns/Schwarzschild recovery construction and negative Yamabe theory already made explicit in I52, the elementary harmonic-map identity (3.1), and Hartman's negative-curvature uniqueness theorem. The core-position obstruction and its actual balanced diagonal are the conclusions proved here. No worldwide priority claim follows from this local derivation.

Appendix A. The exact Schwarzschild donor has no negative gauge mode

The marked obstruction should not be attributed to an unverified negative mode on the handle donor. Its vector Jacobi quadratic form can be evaluated directly.

Write the four-dimensional Riemannian scalar-flat Schwarzschild metric as

$$g_S = \rho^2(dt^2 + \gamma_{S^3}), \qquad \rho = 2a\cosh t, \quad v = \tanh t, \quad \sigma = \operatorname{sech}^2 t.$$

Its coordinate Ricci components are $\mathrm{Ric}_{tt} = -3\sigma$ and $\mathrm{Ric}_{ab} = \sigma\gamma_{ab}$. For a compactly supported vector field set

$$\mathscr{Q}_S(X) = \int (|\nabla X|^2 - \mathrm{Ric}(X,X))\, dV.$$

Decompose angular components by scalar gradients and coclosed vector harmonics on the round unit S^3. For a scalar harmonic Y with $\int Y^2 = 1$, $-\Delta_\gamma Y = \lambda Y$, $\lambda = l(l+2)$, take $X = u(t)Y\partial_t + f(t)\nabla Y$. Integration by parts in the radial first-derivative terms gives

$$\mathscr{Q}_S(X) = \int \rho^4\{u'^2 + \lambda f'^2 + (\lambda + 2\sigma)u^2 + \lambda(\lambda - 4)f^2 - 4\lambda v u f\}\, dt.$$

After $z = \rho^2 u$, $y = \sqrt{\lambda}\rho^2 f$, this is

$$\int \left\{ z'^2 + y'^2 + \left\langle \begin{pmatrix} \lambda + 4 & -2\sqrt{\lambda}\, v \\ -2\sqrt{\lambda}\, v & \lambda - 2\sigma \end{pmatrix} \begin{pmatrix} z \\ y \end{pmatrix}, \begin{pmatrix} z \\ y \end{pmatrix} \right\rangle \right\} dt. \tag{A.1}$$

For $\lambda \geqslant 3$ the matrix minus the identity has nonnegative diagonal and determinant

$$(\lambda - 3)(\lambda + 3 - 2v^2) \geqslant 0.$$

The radial $\lambda = 0$ block is $z'^2 + 4z^2$. For a coclosed vector harmonic with rough eigenvalue μ, the corresponding potential is $\mu + 2 - 2\sigma$. Here $\mu \geqslant 2$: the integrated symmetric-gradient identity on S^3, with divergence zero and Ricci 2γ, gives $\int |\nabla V|^2 \geqslant 2\int |V|^2$. These blocks exhaust the vector fields. Hence

$$\mathscr{Q}_S(X) \geqslant \int \rho^{-2}|X|^2\, dV. \tag{A.2}$$

Since $\mathrm{Ric}(X,X) \leqslant \rho^{-2}|X|^2$, also

$$\mathscr{Q}_S(X) \geqslant \frac{1}{3}\int (|\nabla X|^2 + \rho^{-2}|X|^2)\, dV. \tag{A.3}$$

The inequality extends by cutoff density to this finite-energy space. It excludes a finite-energy kernel or negative mode on the exact donor. It says nothing about a uniform supremum estimate for the strain of a gauge through an entire degenerating marked manifold; Theorem 2 shows why that stronger statement needs different data.